Number Systems: From $\mathbb{N}$ *to* $\mathbb{R}$ ended with the real numbers as a complete ordered field. *Analytic Preliminaries*, Chapter 1 built the complex numbers on those real numbers. This volume starts with the arithmetic of the integers, divisibility and congruences, then takes these results and builds the algebra that linear algebra and the algebra volumes assume on their first page. It studies fields as objects of their own, introduces groups through permutations, makes polynomials into formal objects, and proves that polynomials over a field have the arithmetic of the integers. It ends with the Fundamental Theorem of Algebra: every non-constant complex polynomial has a root.

Who it is for. The reader we have in mind has worked through *Logic and Proof*, *Sets and Functions* and the main line of *Number Systems: From* $\mathbb{N}$ *to* $\mathbb{R}$, and has read the complex numbers of *Analytic Preliminaries*, Chapter 1, or has their habits from elsewhere. That means running an induction, checking that an operation on classes is well defined, and proving a statement about every element of a ring from the ring axioms alone. The chapters point back to those volumes wherever they use a definition or a result, with a link, so a reader who is unsure of a step can look it up. Much of the material is familiar in outline from school or from a first linear algebra course; what is new is that every rule is proved, and that nothing is used before it is proved.

What it covers.

Chapter	Title	What it does
1	Divisibility and Congruences	Bézout's theorem and the Euclidean algorithm, Euclid's lemma and unique prime factorization, ideals and quotient rings, $\mathbb{Z}/n\mathbb{Z}$ as a commutative ring, units, $\mathbb{Z}/p\mathbb{Z}$ a field, and the Chinese remainder theorem (moved from *Number Systems*; in revision)
2	Fields	The field axioms and their consequences; the finite fields $\mathbb{F}_p$; subfields and the subfield test; field homomorphisms; the characteristic and the prime subfield; the binomial theorem, the Frobenius map and Fermat's little theorem
3	Groups, Subgroups, and Homomorphisms	Groups, products of many factors and integer powers; subgroups and the subgroups of $\mathbb{Z}$; homomorphisms, kernels and isomorphisms; permutations, cycle notation and the disjoint cycle decomposition; transpositions generate the symmetric group

Chapter	Title	What it does
4	Formal Polynomials	A polynomial as a sequence of coefficients; the ring $R[x]$ over a commutative ring; degree and integral domains; evaluation and substitution as ring homomorphisms, and why a polynomial is not its function; the formal derivative
5	Polynomials over a Field	Division with remainder, the greatest common divisor and the Euclidean algorithm in $F[x]$; irreducible polynomials and unique factorization; roots, multiplicity and repeated roots; interpolation, the rings $F[x]/\langle m\rangle$ and the Chinese remainder theorem; the Fundamental Theorem of Algebra and the factorization of real polynomials

The complex numbers of *Analytic Preliminaries*, Chapter 1 are the volume's most important example, and Chapter 2 names what they have in common with $\mathbb{Q}$, $\mathbb{R}$ and $\mathbb{Z}/p\mathbb{Z}$. Chapter 3 uses the complex numbers and Chapter 2 only in examples, remarks and one review exercise. Chapters 4 and 5 form a pair. Chapter 4 builds polynomials over any commutative ring and asks only what can be done without dividing; Chapter 5 takes the coefficients in a field and runs again, with the degree in place of size, the arithmetic that Chapter 1 ran in $\mathbb{Z}$.

How to read it. Read the chapters in order; each uses the ones before it. Every section has Quick checks with folded answers; answer each one before you unfold it. The exercises of a section come in three groups, A to check understanding, B for practice and C to go deeper, and the last section of each chapter adds R items that reach back to earlier chapters. A chapter's opening page lists what it needs, with links, and the time it takes. Second-reading folds hold edge cases and longer justifications. They can stay closed on a first reading, but they are not optional, since later proofs may use what they contain.

No borrowed fact. Every result of this volume is proved from the earlier volumes. The complex numbers come from *Analytic Preliminaries*, Chapter 1, which assumes five facts about cosine and sine for polar form and the roots of unity; this volume uses those only in examples and remarks, and Chapter 5 proves the Fundamental Theorem of Algebra from the square roots of *Analytic Preliminaries*, Section 1.4 and the completeness of $\mathbb{R}$ alone. Nothing in the volume depends on *Analysis on the Real Line*.

Time. The chapter openings add up to about 130 hours on the core track and about 185 hours for everything.

What comes next. The linear algebra volumes, beginning with *Spaces and Maps*, take their scalars from a field and use the complex numbers, polynomials and their roots throughout: Chapter 2's fields and subfields, Chapter 4's evaluation of a polynomial, Chapter 5's division

with remainder and the Fundamental Theorem of Algebra, and Chapter 3's permutations, which *Determinants and the Structure of Operators* needs for the determinant. The algebra volumes start where this one stops. *Groups* continues Chapter 3 with cyclic groups, the sign of a permutation, cosets and Lagrange's theorem; *Rings and Fields* continues Chapters 2, 4 and 5 with fraction fields, polynomials over the integers, Gauss's lemma and field extensions. Many of their first definitions are the ones proved here, under the same names.